

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Aquaculture - Set 1

Mark scheme

AQA spec 3.5.2.3; Biological resources; textbook pp. 352-359

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) (1) • LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia (1) • Herbivores are more energy-efficient (lower trophic level) so more food per unit area (1)	3
01.2	<i>1 mark for each point.</i> • Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm (1) • Crossed with normal females (XX) all offspring are XX females (1) • Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently (1)	3
01.3	<i>1 mark for each point.</i> • Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation (1) • Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density (1) • Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification (1) • Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	

Question	Marking guidance	Marks
02.1	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity (up to 2) • Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes (up to 2) • Judgement depends on species, market and site (up to 2)	4
02.2	<i>1 mark for each point.</i> • Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation (1) • Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow (1) • After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg (1)	3
02.3	<i>1 mark for each point.</i> • About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection (1) • Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose (1) • Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) (2) • High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow (2) • Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) (2) • Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes (2)	8
03.2	<i>1 mark for each point.</i> • Predators are killed under licence to protect stock, reducing wild populations of protected / valued species (1) • Better cage designs, netting, fencing, bird scarers exclude predators without killing (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish (1) • Species occupy different niches / trophic levels so they do not compete and all food sources are used (1) • Bottom feeders re-suspend nutrients boosting plankton / plant growth (1) • Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species (1)	4
04.2	<i>1 mark for each point.</i> • IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) (1) • Wastes of one species become nutrients for another so less pollution and extra harvests (1) • Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) (1) • Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar (1)	4
04.3	<i>1 mark for each point.</i> • Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages (1) • Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	

Question	Marking guidance	Marks
05.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
	Model answer - Level 3 (9 marks)	
	Wild catches have not increased since about 1990 because most stocks are fully or over-exploited, while aquaculture has grown rapidly and now approaches the wild catch, so the question is how far it can take over. Its advantages are real: limiting factors, breeding, gender and disease can be controlled, yields per area are high, and fish have low food conversion ratios because they need not support their weight or keep warm; herbivores such as tilapia and carp and filter-feeding molluscs need no fishmeal at all.	

Question	Marking guidance	Marks
	Its limits are equally real. The carnivorous species consumers prefer - salmon, shrimp - are fed fishmeal from wild forage fish, so their farming still depends on fishing and depletes species such as sandeels on which puffins rely; some species, such as bluefin tuna, will not breed in captivity. Farms destroy mangroves, release organic waste, pesticides and antibiotics that breed resistance, spread lice, allow escapes that contaminate wild gene pools and provoke predator control, though siting, closed systems, polyculture and vaccines can reduce these. Meanwhile fishing itself can be sustainable if quotas based on good stock data, gear design, size and effort limits and no-take zones are enforced.	
	Aquaculture can therefore supply most of the growth in demand, especially through herbivorous fish and shellfish, and can relieve pressure on wild stocks, but it cannot fully replace fishing while carnivorous species depend on wild fishmeal and some species cannot be farmed. Both must be managed sustainably; neither is a substitute for the other.	
	Weaker answer - Level 1 (2 marks)	
	Aquaculture is farming fish instead of catching them. Fish farms can produce a lot of fish and they do not run out like wild fish do. But fish farms cause pollution and diseases can spread to wild fish. Salmon farms feed the salmon on fish which have to be caught. So aquaculture can replace some fishing but not all of it.	
	What makes the difference	
	• Level 1 reaches a reasonable conclusion but supports it with two sentences; Level 3 gives the context (flat wild catch since 1990), the advantages in detail and the impacts with ways to reduce them.	
	• Level 3 distinguishes between species types (herbivores and molluscs vs carnivores) - the key to 'to what extent'.	
	• Level 3 also considers whether fishing can be made sustainable, which the comparison requires.	
05.2	<i>1 mark.</i> • A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	